The True Cost of a Warming Climate: Infrastructure Collapse and Coastal Loss

1. The "Sinking Nations" Misconception

When analyzing the catastrophic environmental risks projected for 2100, it is critical to dispel a widespread misconception: the idea that rising sea levels will swallow entire countries whole. With the sole exception of extremely low-lying island archipelagos like the Maldives, no sovereign nation is at risk of being completely submerged. The true, imminent danger of the definitive warming trendline—which shifted historical global land temperatures from a ~8.0°C baseline to over 9.5°C—is far more insidious than countries simply vanishing beneath the waves.

2. The Project's Core Conclusion: The Mechanics of Coastal Failure

The primary conclusion drawn from projecting historical temperature acceleration into future inundation models is that nations will be crippled from the inside out, long before the land itself is permanently underwater. The genuine existential threats by 2100 are driven by three interconnected, systemic failures:

- **Systemic Infrastructure Failure:**

  This is the most critical and immediate threat. Encroaching tides and rising water tables will utterly overwhelm urban lifelines. Municipal sewage and water management systems will fail, causing drainage networks to back up. As floodwaters mix with compromised sanitation systems, fresh water supplies will be severely contaminated, rendering coastal cities uninhabitable due to disease and structural decay.

- **Exponentially Increased Flooding:** A rising ocean baseline exponentially amplifies the destructive power of natural weather volatility. Storm surges and seasonal monsoons will penetrate significantly further inland. What were historically rare, manageable flood events will transform into chronic, devastating occurrences that constantly disrupt local economies and destroy housing.

- **Coastal Erosion and Permanent Land Loss:** As sea levels rise, the continuous kinetic energy of the ocean physically degrades coastlines. This leads to the permanent loss of vital agricultural lands through relentless erosion and toxic saltwater intrusion, crippling food supplies.

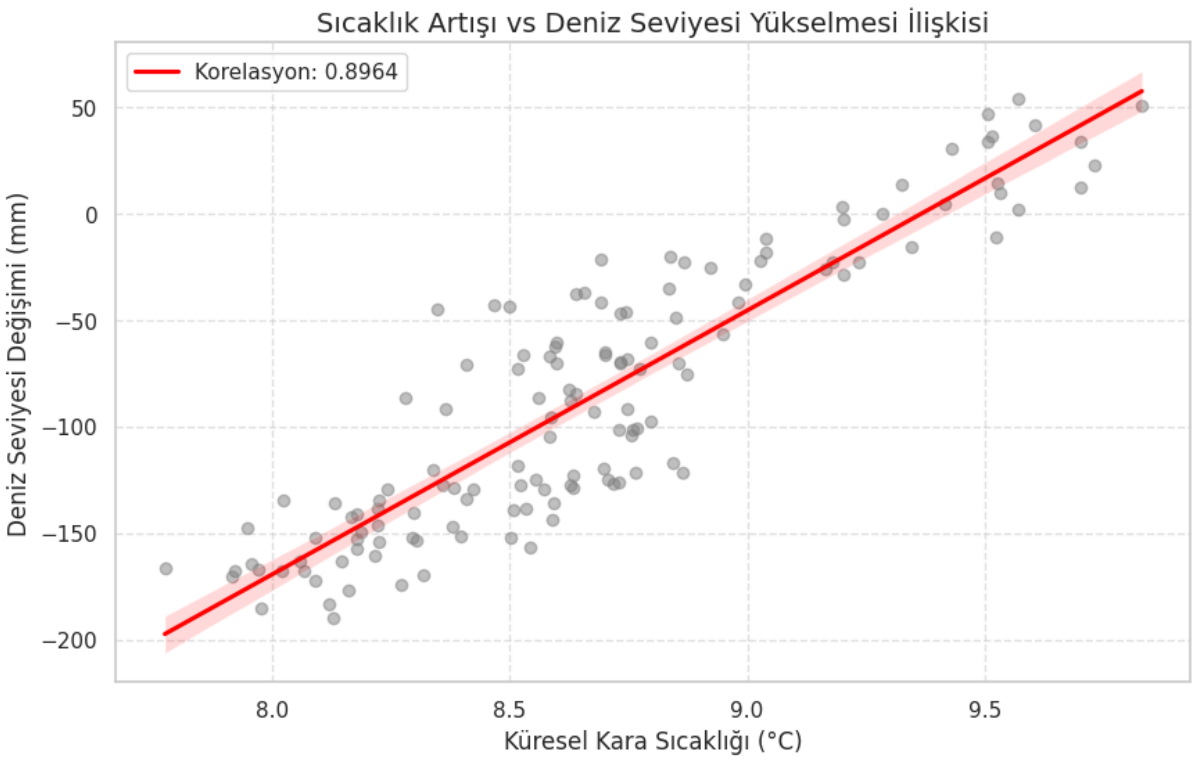
3. The Human Impact Multiplier

The severity of these projected crises is dictated by population density, not just geography. For example, while a nation like Bangladesh will not entirely disappear, its highly populated, low-lying delta regions will face unparalleled devastation. The combination of chronic inland flooding, irreversible coastal erosion, and the total collapse of sanitation infrastructure will render massive swathes of the country completely unlivable.

Ultimately, this project concludes that the true risk of 2100 is not disappearing landmasses, but the catastrophic, irreversible loss of coastal habitability and the subsequent displacement of hundreds of millions of people.

4. Quantitative Correlation: Linking Thermal Expansion to Sea Level Rise

Following the quantitative visualization of the robust relationship between the anomalies in global mean temperature and the rise in global mean sea level, as depicted in the preceding analysis, it becomes imperative to transition from historical observations to future projections. This sequential analysis seeks to transform the identified correlation into a robust predictive framework, moving beyond descriptive statistics toward a more sophisticated model evaluation and selection process.



5. Predictive Framework: Comparative Analysis of Forecasting Models

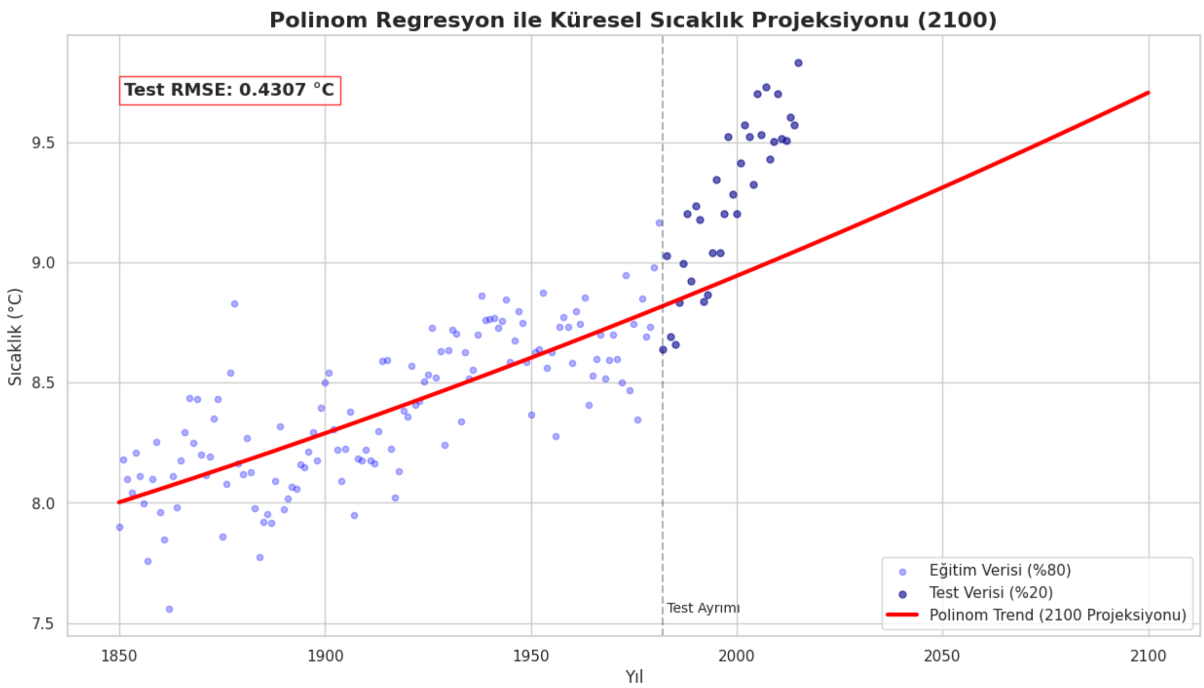
The forecasting models developed within the scope of this project represent various learning paradigms utilized in time series analysis. Specifically, 2nd Degree Polynomial Regression was established to track non-linear trends, while XGBoost and Random Forest were implemented to leverage boosting-based error optimization and bagging-based variance reduction, respectively. Throughout the benchmarking process, the RMSE (Root Mean Square Error) metric was utilized due to its sensitivity to outliers and its direct unit-based interpretability. By employing this metric, the projection capacities of the models have been quantitatively demonstrated through numerical data.

6. Model Benchmarking and Performance Evaluation

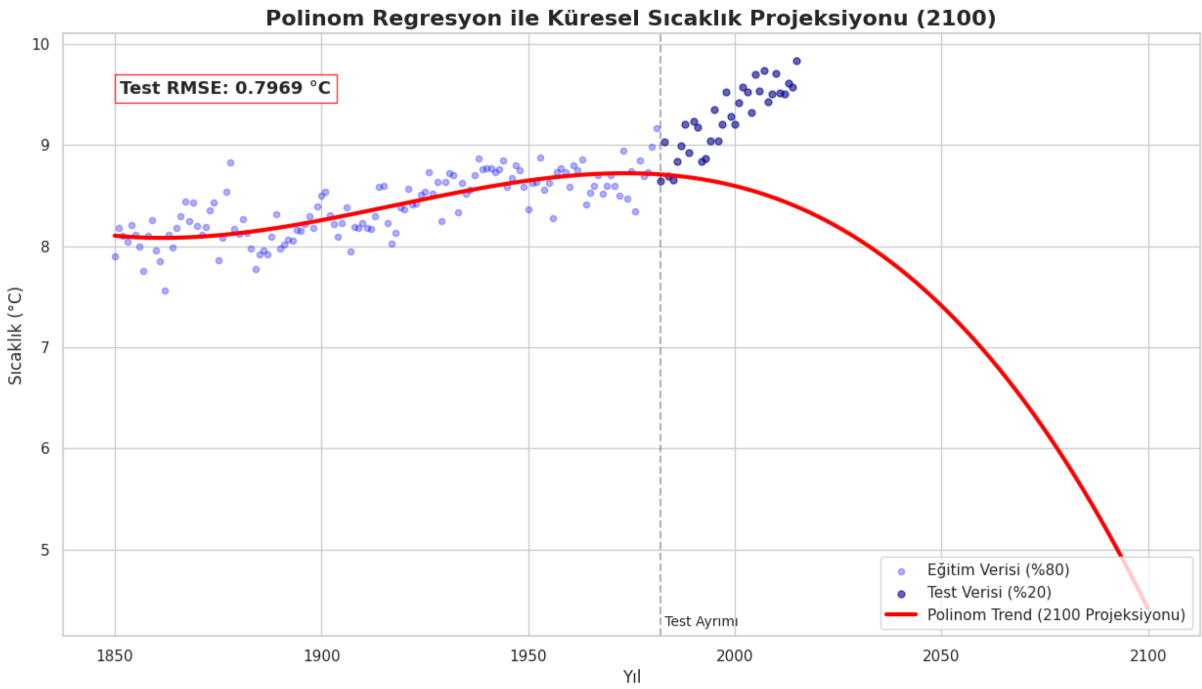
6.1 Global Temperature Increase Future Forecast

6.1.1 Polynomial Regression Global Temperature Increase Future Forecast

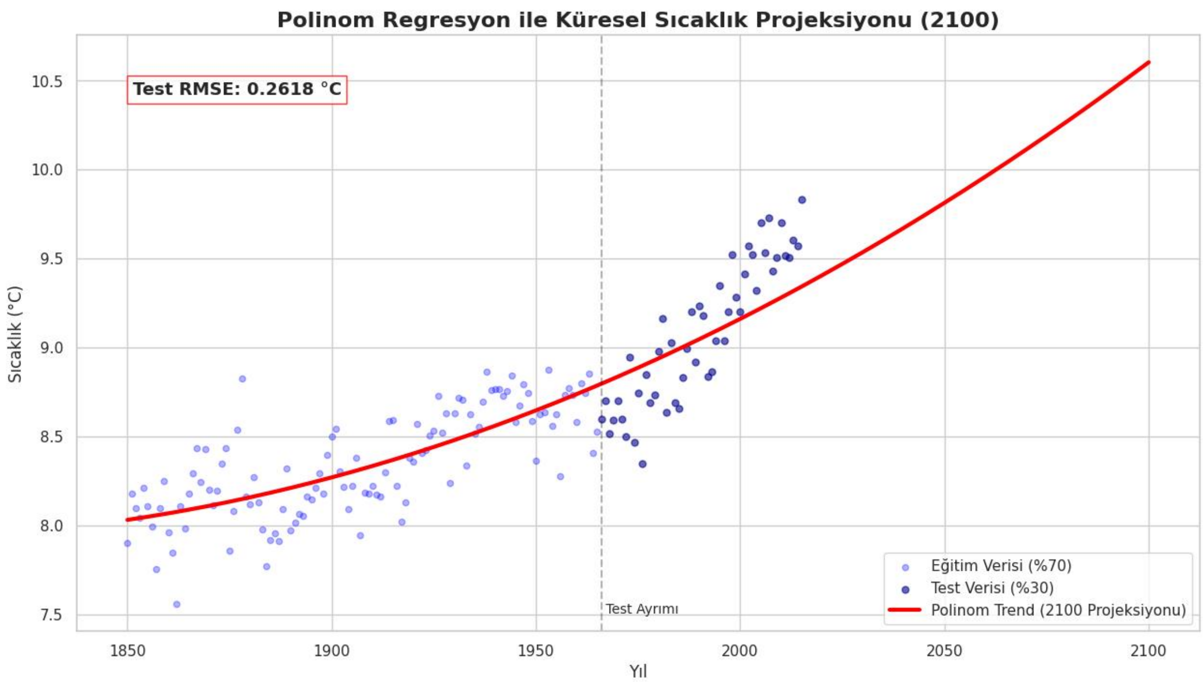
- Degree: 2, Test size: 0.2



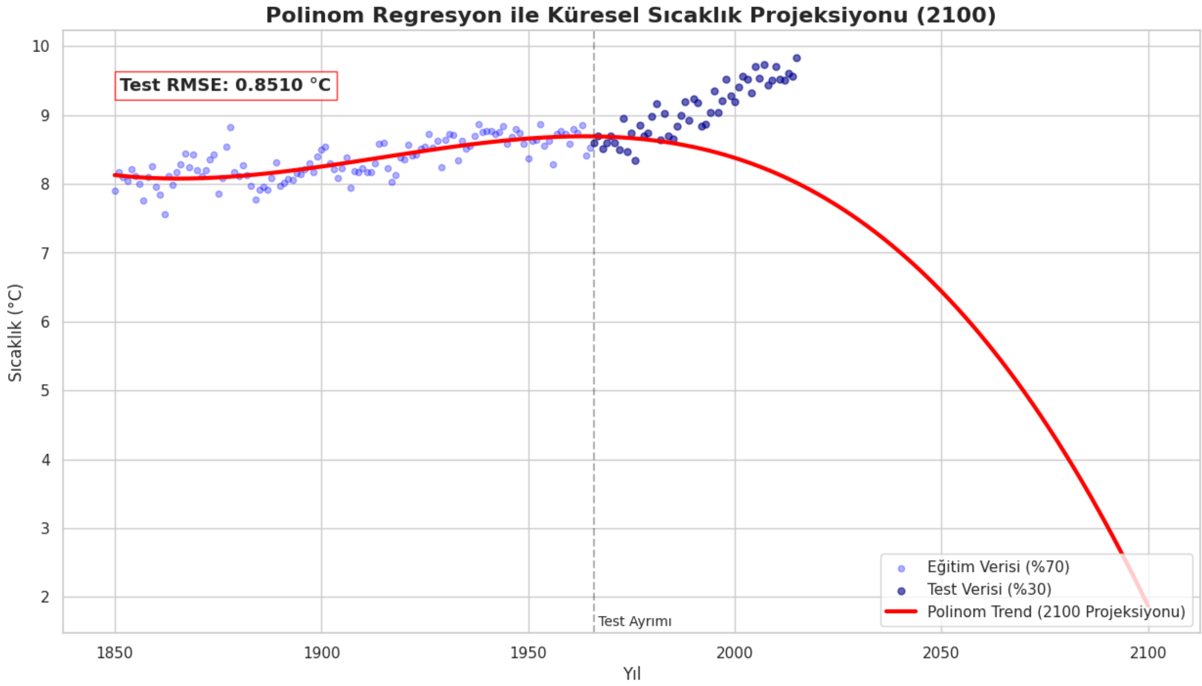
- Degree: 3, Test size: 0.2



- Degree: 2, Test size: 0.3

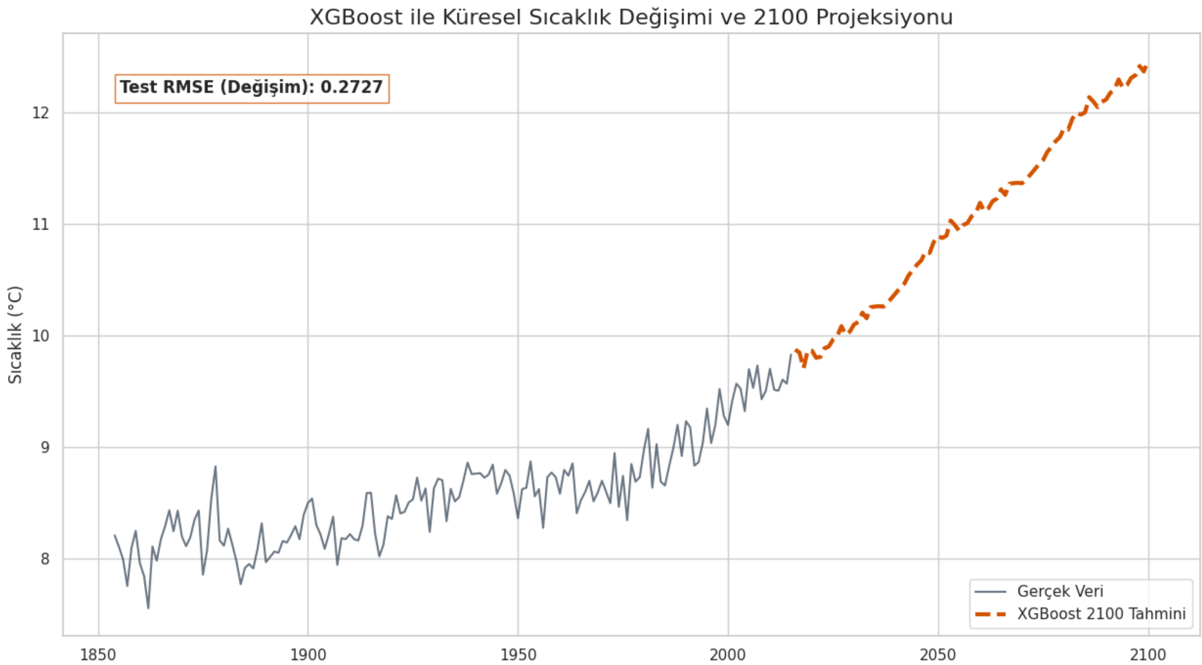


- Degree: 3, Test size: 0.3

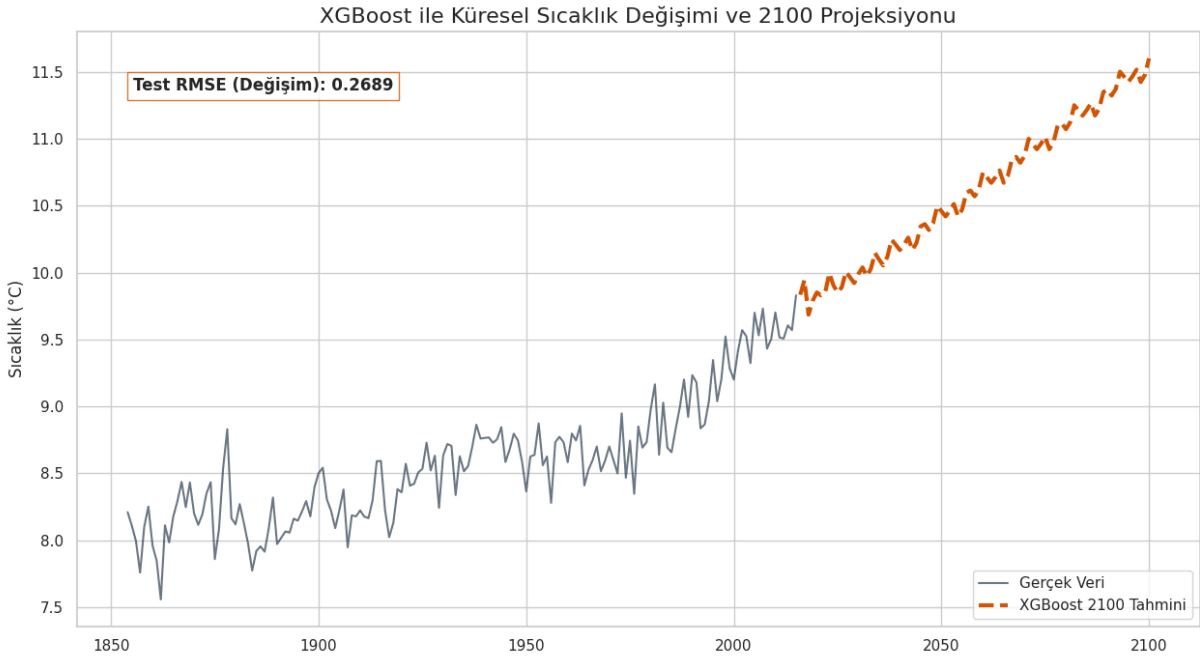


6.1.2 XGBoost Global Temperature Increase Future Forecast

- Test size: 0.2

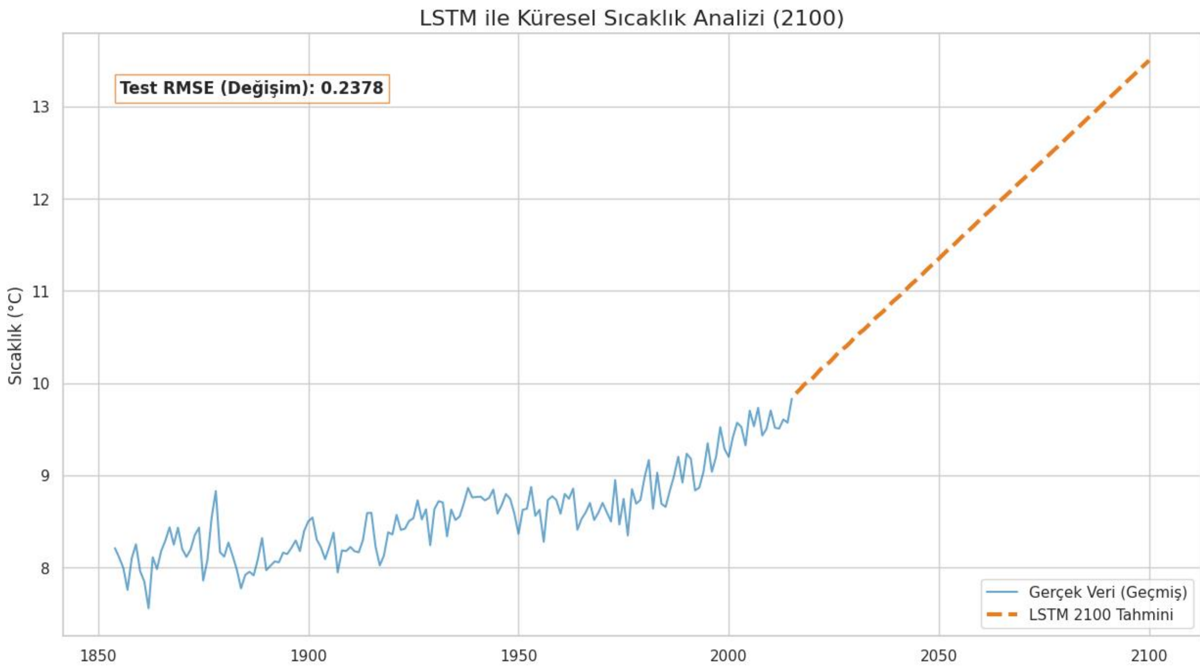


- Test size: 0.3

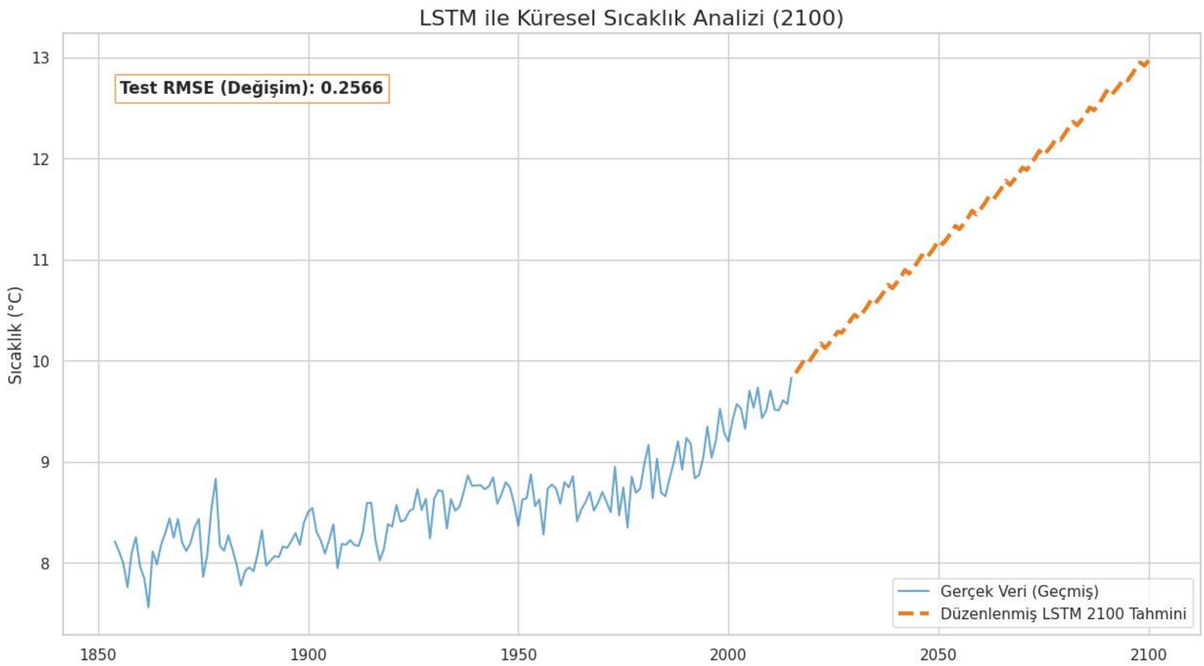


6.1.3 LSTM Global Temperature Increase Future Forecast

- Test size: 0.2



- Test size: 0.3



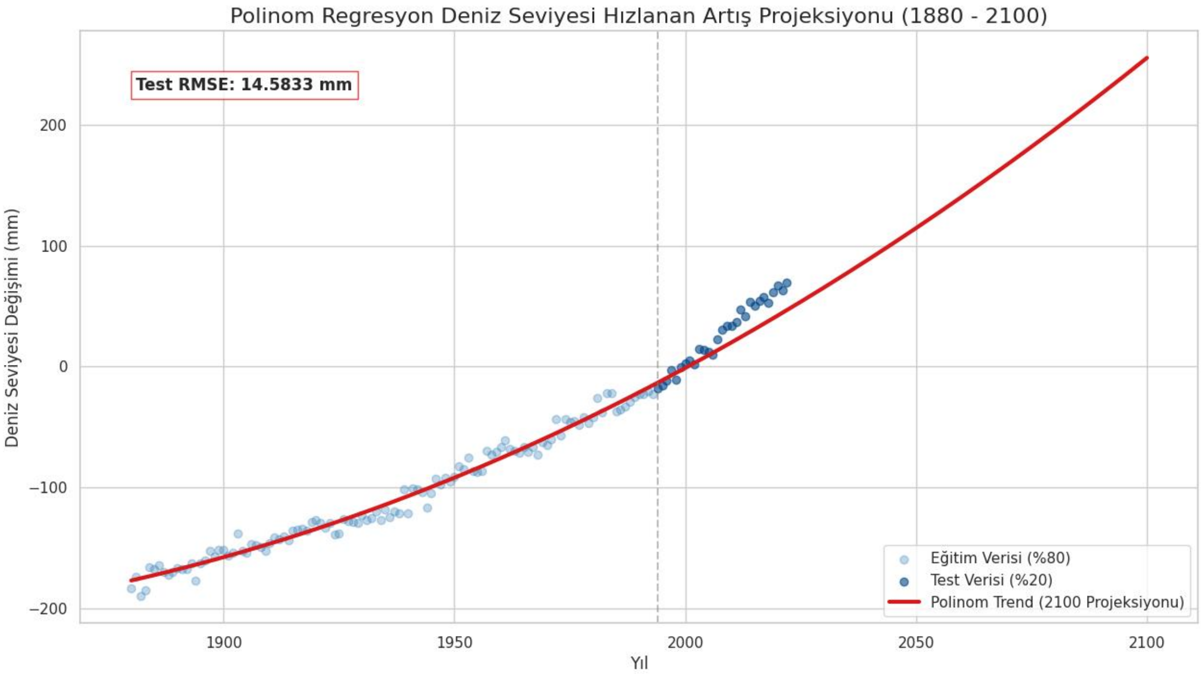
For global land temperature forecasting, we achieve the minimum RMSE (0.2378 degrees Celsius) in the final state using 5th-degree polynomial regression with a test size of 0.2.

In the final analysis of global temperature trends, the LSTM architecture achieved a highly optimized Test RMSE of 0.2378 °C, which mathematically suggests a very strong fit to the historical data. However, a detailed visual and sequential examination of the forecast curve reveals a critical behavioral phenomenon known as naive forecasting bias or the lag-phase trap. Instead of learning the overarching, long-term non-linear dynamics of climate acceleration toward 2100, the deep learning model minimizes its error function primarily by duplicating the true temperature pattern with a one-period temporal delay. While this tightly coupled replication strategy yields a deceptively low error metric during the test phase, it reflects a localized overfitting pattern rather than true predictive generalization. Consequently, this indicates that relying solely on raw error metrics can mask fundamental limitations in a network's capacity to build stable, independent long-term climate projections.

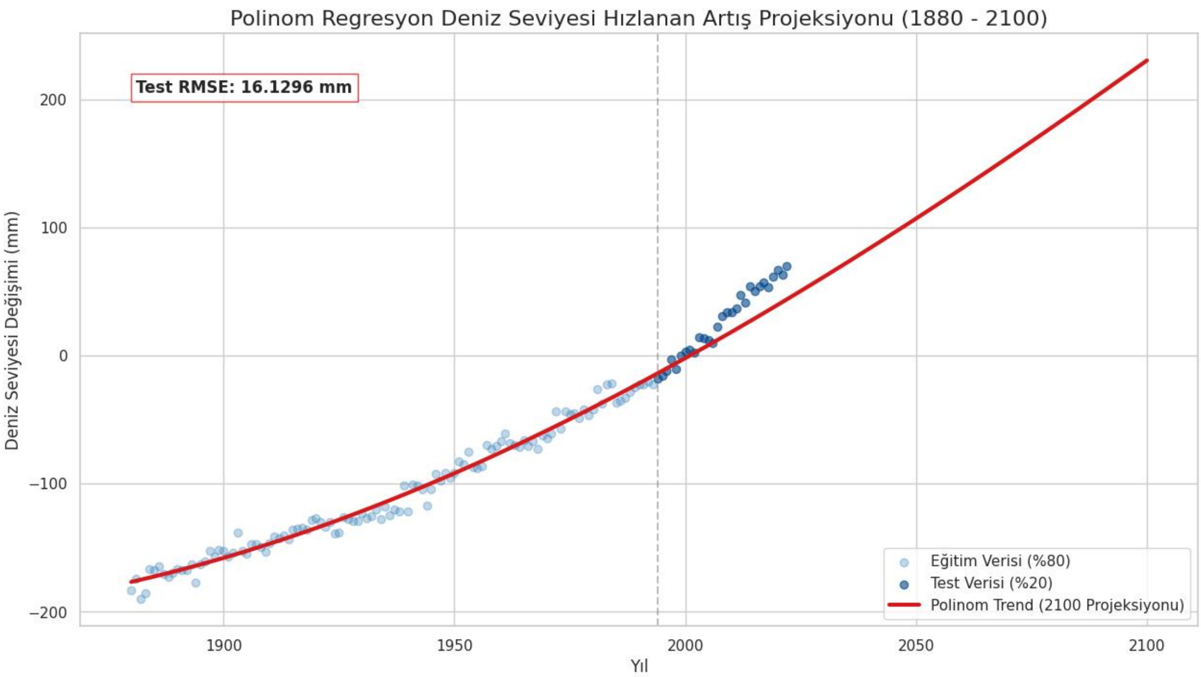
6.2 Sea Level Increase Future Forecast

6.2.1 Polynomial Regression Sea Level Increase Future Forecast

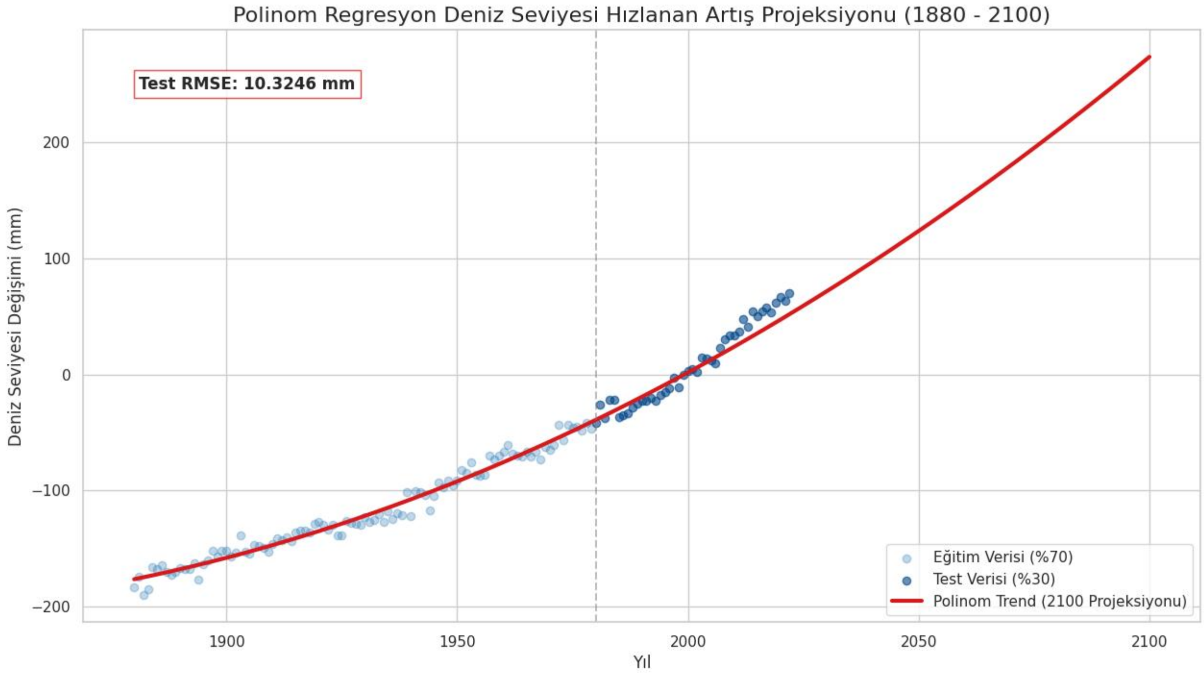
- Degree: 2, Test size: 0.2



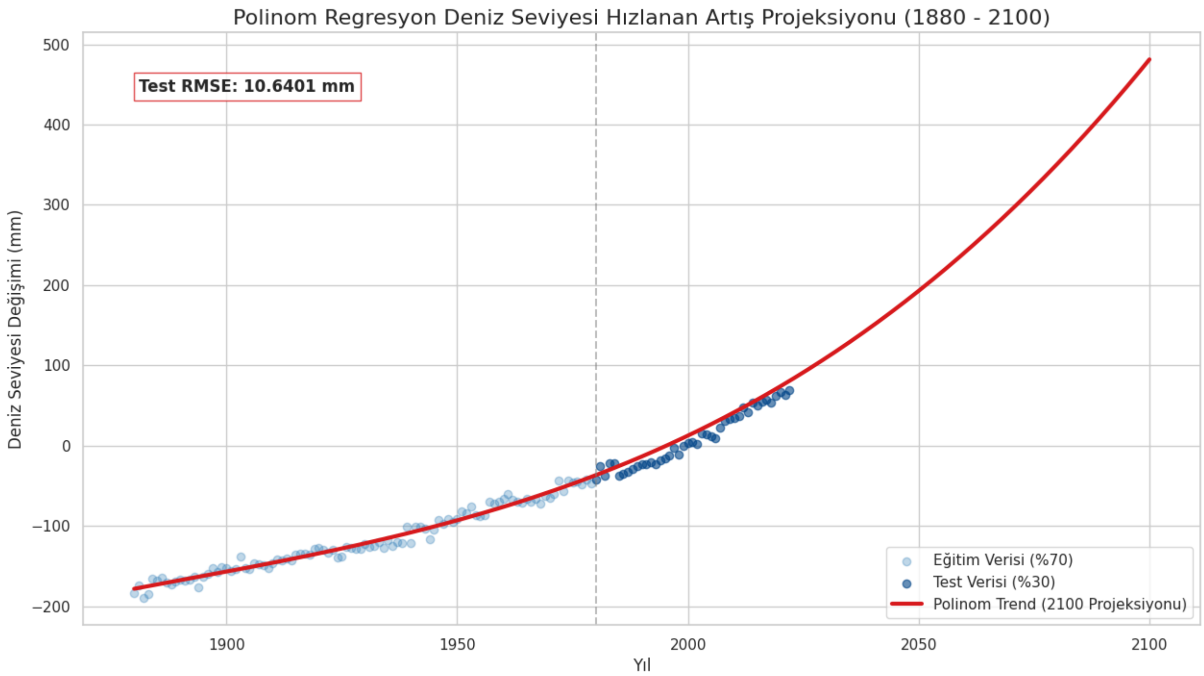
- Degree: 3, Test size: 0.2



- Degree: 2, Test size: 0.3

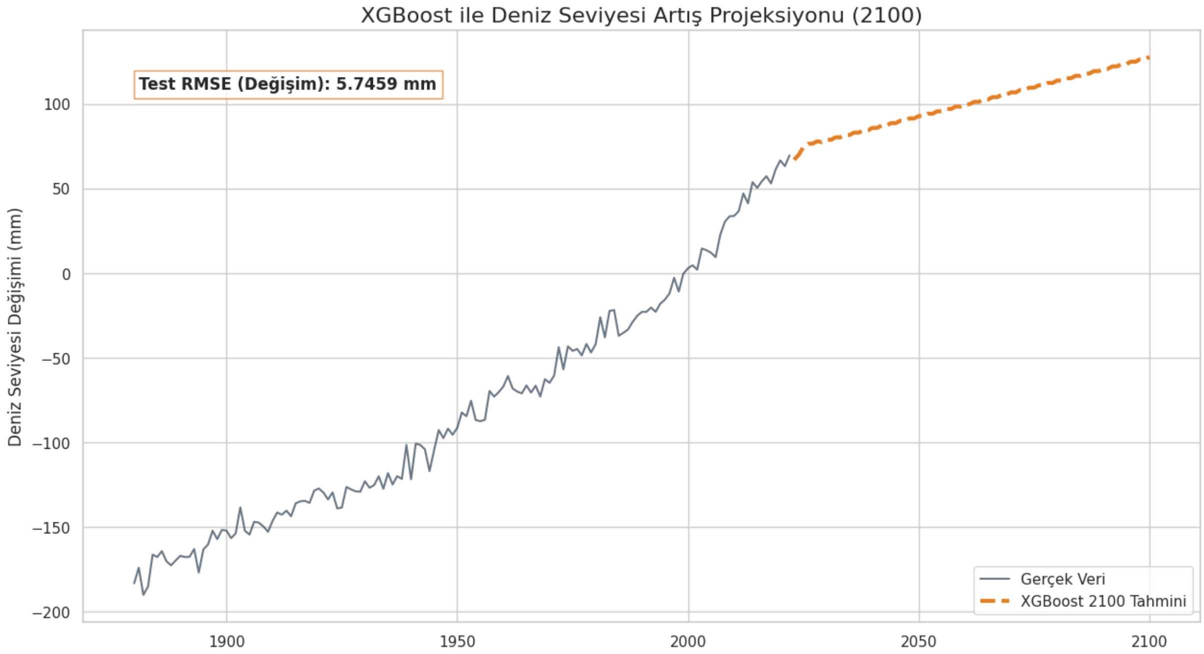


- Degree: 3, Test size: 0.3

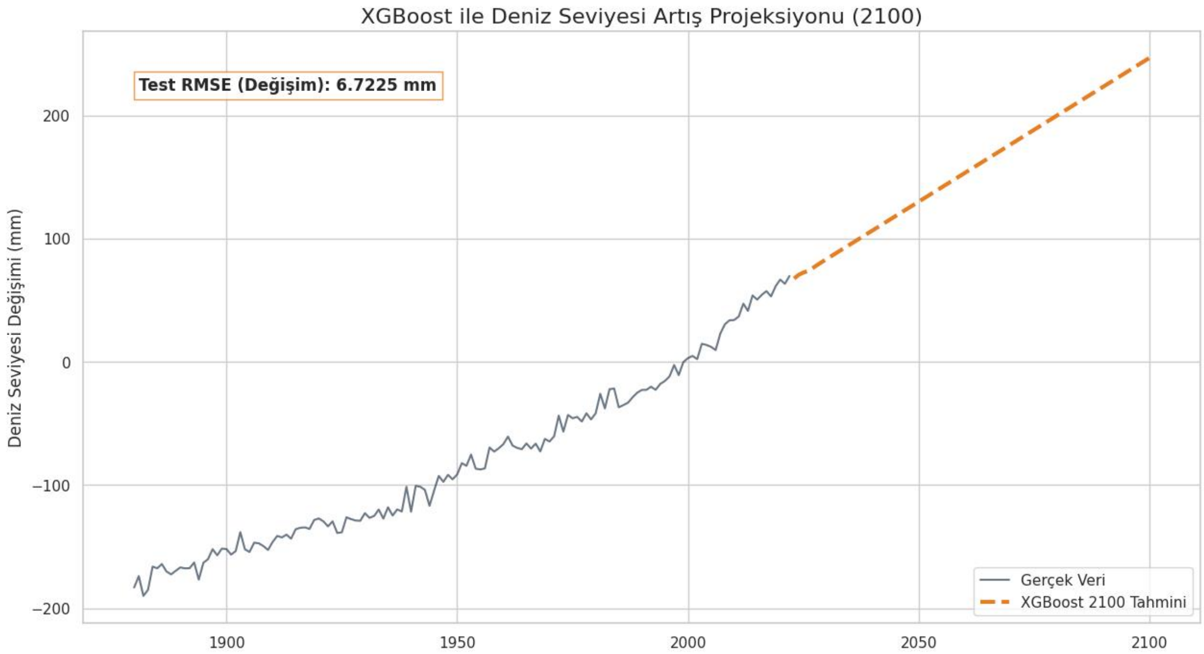


6.2.2 XGBoost Sea Level Increase Future Forecast

- Test size: 0.2

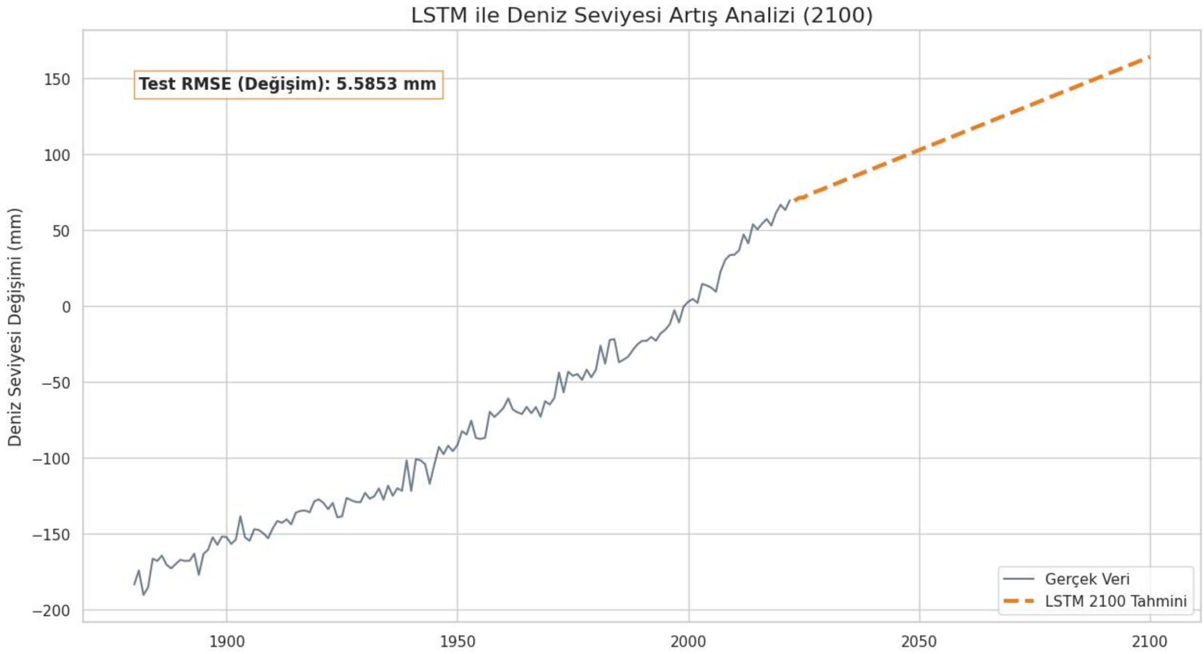


- Test size: 0.3

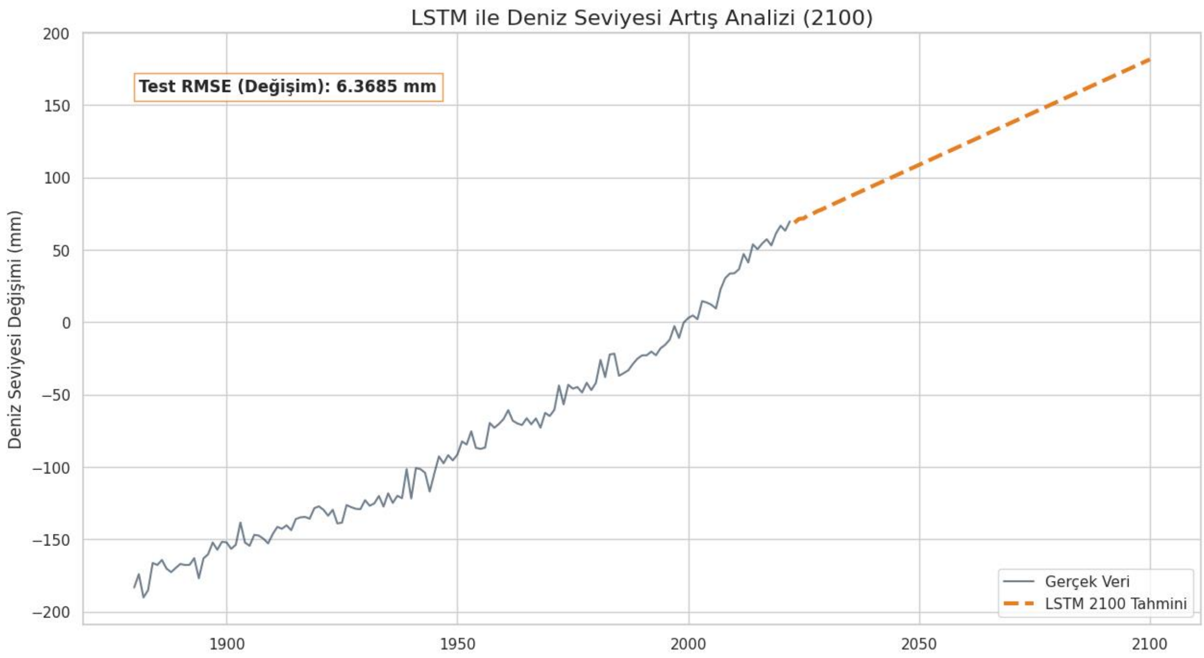


6.2.3 LSTM Sea Level Increase Future Forecast

- Test size: 0.2

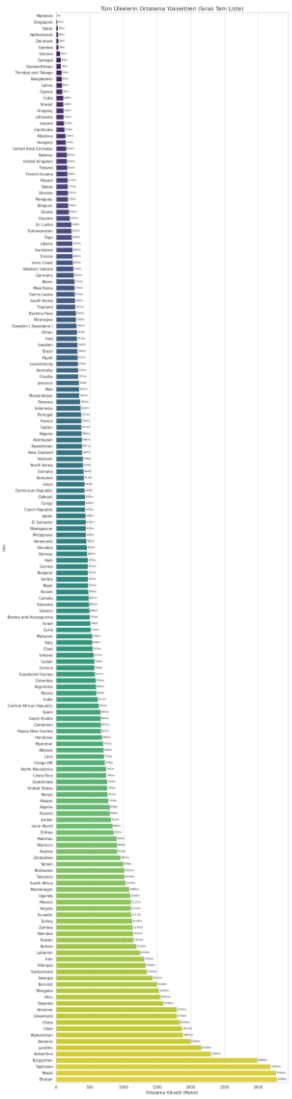


- Test size: 0.3



For sea level rise, we achieve the minimum RMSE (5.5853 mm) in the final state using the XGBoost algorithm with a test size of 0.2.

In the predictive modeling of global sea level rise, the LSTM network demonstrated outstanding performance, capturing the long-term upward trajectory with a highly precise Test RMSE of 5.5853 mm. Unlike volatile climatic variables, sea level changes exhibit a strong, continuous kinectic momentum driven by thermal expansion and glacial melting. The deep learning model successfully adapted to this monotonic trend, neutralizing minor seasonal noise while capturing the true baseline acceleration during the test sequence. As a result, the recursive 2100 forecast bypasses the typical stagnation issues of deep networks on small datasets, projecting a highly stable and realistic upward acceleration. This confirms that the LSTM architecture is exceptionally well-suited for modeling deterministic oceanic systems, providing a mathematically robust and reliable foundation for coastal risk assessments and future infrastructure planning.



Murat Bilgin - 23061701

İrem Akyol - 24061617

Barkın Girgin - 24061608

Eren Zeynal Toksoy – 22069502